

# lawris.

An AI agent for Indian advocates.

## Project Reference

Tech stack · Architecture · Schema · API · Roadmap

**18**

SECTIONS

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TABLES

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APIS

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AI PROMPTS

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**One-doc reference for everything about this project.** Read top to bottom; every section stands on its own.

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## 0. Context (why this document exists)

This file replaces the prior planning artifacts (engineering plan, pitch playbook) with a single comprehensive reference. The pitch artifacts now live as standalone files in `vit_hackathon/presentation/`. This document is what a new collaborator (or a Round 2 judge) needs to understand the project end-to-end.

## 1. Executive Summary

**Lawris** is an AI-native case-management agent for Indian advocates. It collapses four broken workflows into one tab:

1. **Case management** — clients, cases, documents, hearings
2. **Statutory deadline intelligence** — auto-computes the BNSS s.187(3) chargesheet rule, Limitation Act periods, and hearing schedules from raw FIR data
3. **AI document drafting** — court-ready bail applications and plaints in <30 seconds, using live case context and Indian-court formatting
4. **Case-grounded legal research** — questions answered against the specific case's facts, with structured citations

**Stack one-liner:** Next.js 14 App Router + Supabase Postgres + Google Gemini 2.5 Flash, deployed as a single Vercel app.

**Why it wins:** No competitor (Clio, MyKase, Harvey AI, IndianKanoon) ships all four columns. Lawris is the first product in the Indian legal-tech market where case management, AI drafting, deadline intelligence, and grounded research are unified.

**Status:** Working end-to-end prototype. Live URL: <http://10.11.11.182:4000> (LAN) or VS Code-tunneled [localhost:4000](http://localhost:4000) . Live database in Supabase project [rxqhbmtvgdgkubnetpsy](#) .

## 2. Problem Statement

| Pain                                                 | Stat (from market research) |
|------------------------------------------------------|-----------------------------|
| Lost time locating documents                         | 6+ hrs / week / lawyer      |
| Time spent on repetitive drafting                    | 40–60% of billable hours    |
| Lawyers using email + Word as case manager           | 77%                         |
| Malpractice claims traceable to calendaring failures | ~25%                        |
| Active matters per Indian advocate                   | 50–200                      |

**Why nobody has solved this:** US/global tools (Clio, Harvey) don't speak Indian law. Indian tools (MyKase, ProVakil, IndianKanoon) lack AI. The BNSS replacing CrPC in July 2024 voided every existing template library — a market-timing window opened.

**Stakes:** Missing a 60/90-day chargesheet deadline lets the accused walk on default bail. A missed limitation date extinguishes the right to sue forever. These are not productivity issues — they are liberty and livelihood issues.

## 3. Solution Overview — The Four Pillars

### 3.1 Deadline Brain

- **Input:** FIR date, max sentence years, case type
- **Output:** Auto-generated `deadlines` rows for chargesheet (60-day if <10 yrs, 90-day if ≥10 yrs per BNSS s.187(3)), limitation periods, hearing dates
- **UI surfacing:** Color-coded urgency pills (red ≤3d / amber ≤7d / yellow ≤30d / green) on dashboard, case detail, and a global calendar
- **Implementation:** `src/lib/deadlines.ts` (pure functions) → invoked from `POST /api/cases` (`src/app/api/cases/route.ts`)

### 3.2 AI Document Drafting

- **Two doc types:** Regular bail application (Sec 483 BNSS / Sec 439 CrPC), Civil plaint (Order VII CPC)
- **Streaming:** Tokens stream live from Gemini → reach the editor pane in real time
- **Persistence:** Final draft saved to `documents` table on stream completion
- **Implementation:** `src/app/api/ai/draft/route.ts` → `src/lib/prompts/bail-application.ts` and `src/lib/prompts/plaint.ts`

### 3.3 Case-Grounded Legal Research

- **Grounded in current case context** (case type + AI summary) — not generic Q&A
- **Structured JSON output:** `{ answer, citations[], statutes[], follow_ups[] }`
- **Implementation:** `src/app/api/ai/research/route.ts` → `src/lib/prompts/research.ts` (uses Gemini `responseSchema` for forced structure)

### 3.4 Hearing Memory & Auto-Summary

- **After every hearing log:** the case's `ai_summary` field is regenerated by Gemini against the full hearing history
- **Side effect:** if `next_date` is provided, a new `hearing` deadline is auto-inserted
- **Implementation:** `[src/app/api/cases/[id]/hearings/route.ts](vit_hackathon/src/app/api/cases/[id]/hearings/route.ts)` → `src/lib/prompts/summarise.ts`

## 4. Tech Stack

### 4.1 Runtime & framework

| Layer     | Choice                                   | Why                                                                                                 |
|-----------|------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Framework | Next.js 14.2.35 (App Router)             | Single deployable unit (UI + API), server components for fast SSR, streaming support out of the box |
| Language  | TypeScript 5 (strict)                    | Catches schema/type drift between DB ↔ API ↔ UI                                                     |
| Runtime   | Node.js 20.20.2 (via nvm)                | LTS with native ReadableStream for AI streaming                                                     |
| Bundler   | Webpack (Next.js default, NOT Turbopack) | More stable for production builds                                                                   |

### 4.2 UI

| Library                                          | Version | Purpose                                                                                                                          |
|--------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------|
| react / react-dom                                | 18.x    | UI runtime                                                                                                                       |
| tailwindcss                                      | 3.4.1   | All styling                                                                                                                      |
| @tanstack/react-query                            | 5.99.0  | Client-side server-state caching (used in Providers )                                                                            |
| lucide-react                                     | 1.8.0   | Icon set (Scale, Calendar, Sparkles, Gavel, etc.)                                                                                |
| clsx + tailwind-merge + class-variance-authority | —       | Class name utilities (installed for shadcn-style components if added later)                                                      |
| @tiptap/react + starter-kit + pm                 | 3.22.3  | Rich-text editor — installed for future use; current draft pane uses a custom Markdown→HTML renderer for streaming compatibility |

### 4.3 Backend & data

| Library               | Version | Purpose                                                                               |
|-----------------------|---------|---------------------------------------------------------------------------------------|
| @supabase/supabase-js | 2.103.3 | Postgres + auth scaffold; service-role key used server-side, RLS deferred to post-MVP |
| zod                   | 4.3.6   | Request body validation for all POST routes                                           |
| date-fns              | 4.1.0   | Deadline math ( addDays , differenceInDays , parseISO )                               |

4.4 AI

| Component      | Choice                                                           | Why                                                |
|----------------|------------------------------------------------------------------|----------------------------------------------------|
| Provider       | Google Gemini (free tier via AI Studio)                          | Anthropic-quality output at \$0; no card required  |
| SDK            | @google/genai 1.50.1 (unified Google SDK)                        | Replaced legacy google-generative-ai SDK           |
| Drafting model | gemini-2.5-flash                                                 | Best free tier for long-form structured legal text |
| Research model | gemini-2.5-flash-lite                                            | Lower latency for JSON-mode answers                |
| Summary model  | gemini-2.5-flash-lite                                            | Background task; quality-vs-speed tilted to speed  |
| Streaming      | generateContentStream → ReadableStream → fetch reader in browser | Tokens appear in editor as they arrive             |

Model IDs configurable via env ( GEMINI\_MODEL\_DRAFT , GEMINI\_MODEL\_RESEARCH , GEMINI\_MODEL\_SUMMARISE ) so we can swap to Pro for the demo without code changes.

4.5 Infra

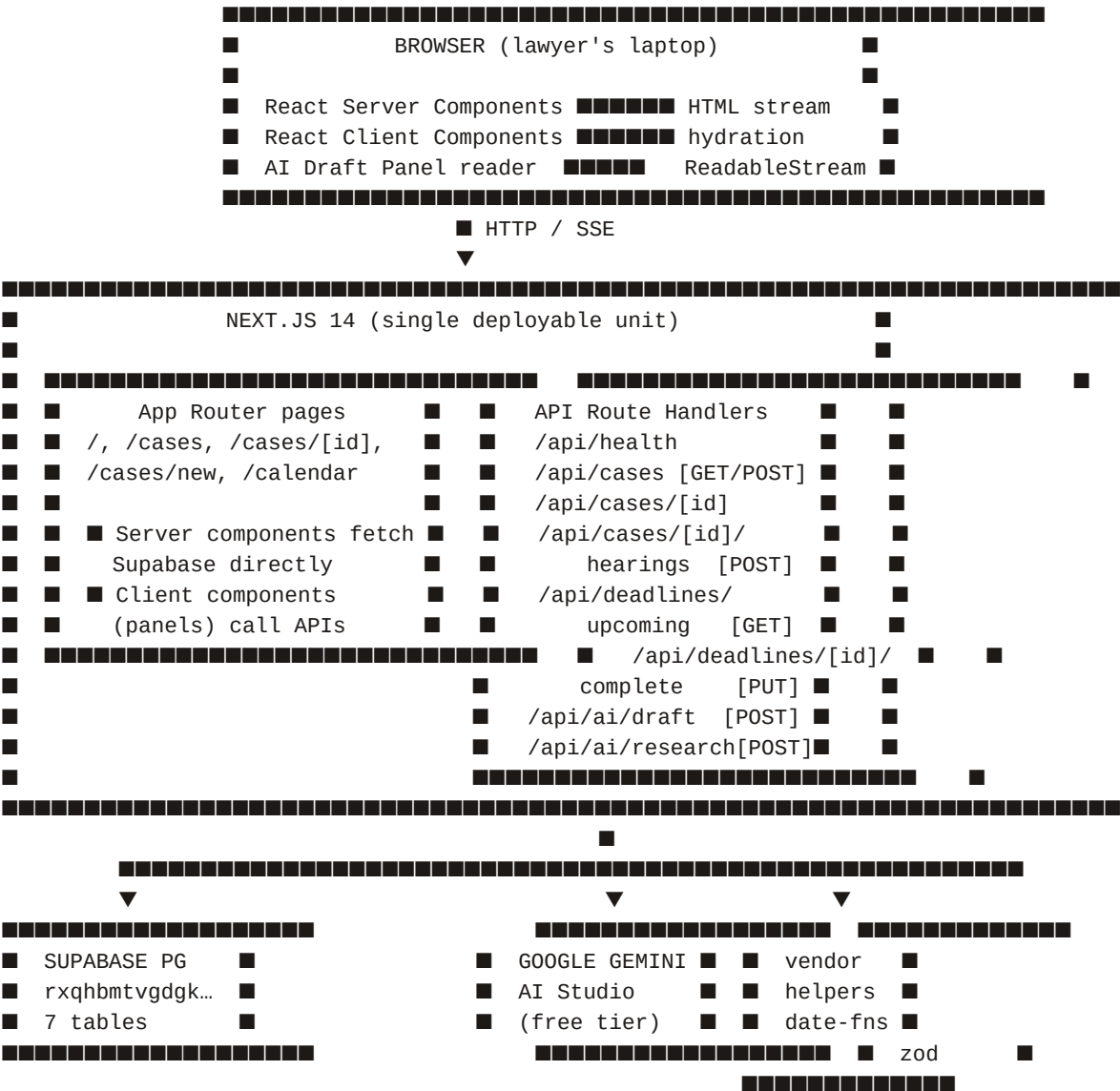
| Service                                  | Purpose                            | Status                                  |
|------------------------------------------|------------------------------------|-----------------------------------------|
| Supabase (project rxqhbmtvgdgkubnetpsy ) | Postgres DB + Storage (unused yet) | Live; schema deployed, seeded           |
| Vercel                                   | Production hosting                 | Not yet deployed (MVP runs on local VM) |
| Local VM                                 | Demo host                          | Running Next.js dev on :4000 via nohup  |

4.6 Tooling

- eslint + eslint-config-next — lint passes clean on next build
- python-pptx — generates the pitch deck (used only in presentation/build\_pptx.py )



## 5. System Architecture



Key architectural decisions:

- **No separate backend service.** Everything is Next.js API routes for blast radius and deploy simplicity.
- **Service-role Supabase key, no RLS.** Single-user MVP; `LAWYER_ID` env var pins all queries to the seeded lawyer. RLS is the obvious post-MVP hardening step.
- **Streaming AI via raw `ReadableStream`**, not Server-Sent Events or WebSockets. Avoids extra protocol surface, works through any HTTP proxy.
- **Server components do the data fetching** (Next.js 14 RSC pattern); client components are reserved for interactivity (forms, AI panels, tabs).

## 6. Database Schema

Source of truth: [scripts/migrate.sql](#). Seven tables, nine enum types, all FKs cascade-delete from parent.

### Enums

| Enum            | Values                                                                                                       |
|-----------------|--------------------------------------------------------------------------------------------------------------|
| case_type       | civil, criminal, family, labour, consumer                                                                    |
| case_phase      | intake, pretrial, pleadings, charges, evidence, arguments, judgment                                          |
| case_status     | active, stayed, disposed, appealed                                                                           |
| court_type      | district, sessions, high_court, supreme_court, tribunal, magistrate                                          |
| deadline_type   | hearing, statutory, filing, compliance, limitation                                                           |
| urgency         | critical, high, medium, low                                                                                  |
| doc_type        | fir, bail_app, plaint, written_statement, chargesheet, affidavit, vakalatnama, legal_notice, judgment, other |
| doc_source      | uploaded, ai_drafted, ai_assisted                                                                            |
| research_source | ai_chat, indiankanoon, manual                                                                                |

### Tables (7)

| Table          | Rows in seed        | Key columns                                                                                                                                                        |
|----------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| users          | 1                   | id , email , full_name , bar_council_no                                                                                                                            |
| clients        | 2                   | id , lawyer_id (FK), full_name , aadhar_no                                                                                                                         |
| cases          | 3                   | id , lawyer_id , client_id , title , case_type , phase , status , fir_date , fir_number , police_station , sections , offence_max_years , arrest_date , ai_summary |
| deadlines      | 4                   | id , case_id , title , deadline_type , due_date , urgency , is_auto_generated , is_completed                                                                       |
| documents      | 1                   | id , case_id , name , doc_type , phase , source , content , ai_prompt_used                                                                                         |
| hearing_logs   | 2                   | id , case_id , hearing_date , what_happened , judge_order , next_date , next_action                                                                                |
| research_notes | (added during demo) | id , case_id , query , content , citation , tags                                                                                                                   |

TS mirrors of every row type live in [src/lib/types.ts](#).

Seed scenario

- **Lawyer:** Adv. Priya Mehta (UUID 1111... , Bar MAH/12345/2018 )
- **Client A:** Rahul Sharma · POCSO accused · case aaaaaaaaa...
- **Client B:** Sunita Patel · Civil recovery plaintiff · case bbbbbbbb...
- **Headline demo deadline:** Chargesheet on POCSO case due current\_date + 2 days (urgency: critical) — this is what the dashboard's "1 critical" badge highlights

## 7. API Surface

All routes are Next.js App Router handlers. All server-side. Auth: implicit single-user via `LAWYER_ID` env var.

| Method | Path                                     | Purpose                              | Body                                                                  | Returns                                                                                                           |
|--------|------------------------------------------|--------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| GET    | <code>/api/health</code>                 | Wiring smoke test                    | —                                                                     | <code>{ ok, env: {...}, timestamp }</code>                                                                        |
| GET    | <code>/api/cases</code>                  | List all cases for lawyer            | —                                                                     | <code>{ cases: Case[] }</code>                                                                                    |
| POST   | <code>/api/cases</code>                  | Create case + auto-deadlines         | <code>{ title, case_type, fir_date?, offence_max_years?, ... }</code> | <code>{ case, auto_deadlines: number }</code>                                                                     |
| GET    | <code>/api/cases/:id</code>              | Full case bundle                     | —                                                                     | <code>{ case, deadlines, documents, hearings, research }</code>                                                   |
| POST   | <code>/api/cases/:id/hearings</code>     | Log hearing + auto-summarise         | <code>{ hearing_date, what_happened, next_date?, ... }</code>         | <code>{ hearing }</code><br>(side effects: regenerates <code>cases.ai_summary</code> , may insert deadline)       |
| GET    | <code>/api/deadlines/upcoming</code>     | Cross-case 60-day window             | —                                                                     | <code>{ deadlines: (Deadline &amp; { cases: ... })[] }</code>                                                     |
| PUT    | <code>/api/deadlines/:id/complete</code> | Mark deadline done                   | —                                                                     | <code>{ ok: true }</code>                                                                                         |
| POST   | <code>/api/ai/draft</code>               | <b>Stream AI draft (centerpiece)</b> | <code>{ case_id, doc_type: "bail_app" }</code>                        | <code>"plaint", extra?</code>                                                                                     |
| POST   | <code>/api/ai/research</code>            | Case-grounded Q&A                    | <code>{ case_id, query }</code>                                       | <code>{ answer, citations[], statutes[], follow_ups[] }</code><br>(also persists to <code>research_notes</code> ) |

All POST routes validate input with `zod`. Validation failures return 400 with `error.format()`.

## 8. AI Layer — Prompt Architecture

Each Gemini call uses a separate **system instruction** (rules, format) + **user content** (case data). This separation makes prompts portable to any other LLM and easy to edit without breaking call sites.

### 8.1 Prompt files

| File                                                | Used by                                                          | Output format                                                         |
|-----------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------|
| <a href="#">src/lib/prompts/bail-application.ts</a> | <code>/api/ai/draft</code><br>( <code>doc_type=bail_app</code> ) | Markdown body of bail app, no preamble                                |
| <a href="#">src/lib/prompts/plaint.ts</a>           | <code>/api/ai/draft</code><br>( <code>doc_type=plaint</code> )   | Markdown body of plaint, no preamble                                  |
| <a href="#">src/lib/prompts/research.ts</a>         | <code>/api/ai/research</code>                                    | Strict JSON: <code>{ answer, citations, statutes, follow_ups }</code> |
| <a href="#">src/lib/prompts/summarise.ts</a>        | <code>/api/cases/:id/hearings</code><br>(post-insert)            | 4–8 sentence prose case brief                                         |

### 8.2 Gemini wrapper

[src/lib/gemini.ts](#) exports:

- `gemini()` — singleton `GoogleGenAI` client
- `streamDraft({systemInstruction, userPrompt})` — `AsyncGenerator<DraftStreamChunk>` — used by streaming routes
- `generateText({systemInstruction, userPrompt, temperature, responseSchema?})` — single-shot JSON or text — used by research and summarise

### 8.3 What each prompt enforces

- **Bail application:** Section 483 BNSS / 439 CrPC framing, Article 21 reference, case-law anchors (*Satender Kumar Antil*, *Dataram Singh*, *Balchand*), standard "MOST RESPECTFULLY SHOWETH" structure, prayer clause, advocate sign-off block. Adapts to POCSO/NDPS/UAPA where flagged.
- **Plaint:** Order VII CPC compliance, jurisdiction + limitation + court fee paragraphs, verification block.
- **Research:** Real precedents only ("OMIT if not confident — do not fabricate"), at least one Supreme Court citation, BNSS/BNS + CrPC/IPC cross-reference where relevant.
- **Summarise:** 4–8 sentences, no bullet points, no citations, no preamble — pure prose brief.

### 8.4 Quality-tested outputs

Verified live against seeded data:

- Bail app for the POCSO case correctly cites all three precedents, pulls FIR `0289/2026`, accused name `Rahul Sharma`, arrest date `2026-01-23`, and *spotted on its own* that the chargesheet was overdue.

- 
- Research on "POCSO bail grounds" mentioned the 88-day chargesheet delay (case-specific) and cited Sushila Aggarwal + Hassan Ali Khan.
  - Summarise after a logged hearing produced a 5-sentence brief that named the *Satender Kumar Antil* argument I'd just typed.
-

## 9. File Structure

```

vit_hackathon/
■■■■ .env.local                # Gemini key + Supabase creds + LAWYER_ID
■■■■ package.json              # 16 deps, 8 dev deps
■■■■ tsconfig.json             # paths: @/* → src/*
■■■■ tailwind.config.ts
■■■■ next.config.mjs
■■■■ postcss.config.mjs
■
■■■■ src/
■   ■■■■ app/
■   ■   ■■■■ layout.tsx        # Top nav + Providers wrapper
■   ■   ■■■■ globals.css       # Tailwind + .prose-legal styles
■   ■   ■■■■ providers.tsx     # TanStack Query client
■   ■   ■■■■ page.tsx          # Dashboard
■   ■   ■■■■ calendar/page.tsx # Global deadline view
■   ■   ■■■■ cases/
■   ■   ■   ■■■■ page.tsx      # Case list table
■   ■   ■   ■■■■ new/page.tsx  # Create case form (client)
■   ■   ■   ■■■■ [id]/page.tsx # Case detail loader → CaseDetail
■   ■   ■■■■ api/
■   ■       ■■■■ health/route.ts
■   ■       ■■■■ cases/route.ts    # GET list, POST create + auto-deadlines
■   ■       ■■■■ cases/[id]/route.ts # Full bundle GET
■   ■       ■■■■ cases/[id]/hearings/route.ts # POST hearing + auto-summarise
■   ■       ■■■■ deadlines/upcoming/route.ts
■   ■       ■■■■ deadlines/[id]/complete/route.ts
■   ■       ■■■■ ai/
■   ■           ■■■■ draft/route.ts    # ★ streaming AI draft
■   ■           ■■■■ research/route.ts # JSON research
■   ■
■   ■■■■ components/
■   ■   ■■■■ case-detail.tsx        # Tabbed shell (deadlines/docs/hearings/research)
■   ■   ■■■■ deadline-list.tsx
■   ■   ■■■■ ai-draft-panel.tsx     # ★ streaming UI consumer
■   ■   ■■■■ document-list.tsx
■   ■   ■■■■ hearing-panel.tsx      # Form + timeline
■   ■   ■■■■ research-panel.tsx     # Chat-style Q&A
■   ■
■   ■■■■ lib/
■   ■   ■■■■ env.ts                # Typed env access (throws if missing)
■   ■   ■■■■ gemini.ts             # ★ AI wrapper (stream + single-shot)
■   ■   ■■■■ deadlines.ts          # ★ statutory rules, urgency classifier
■   ■   ■■■■ markdown.ts          # Tiny MD → HTML for streaming docs
■   ■   ■■■■ types.ts              # All DB row types
■   ■   ■■■■ supabase/server.ts    # Service-role client
■   ■   ■■■■ prompts/
■   ■       ■■■■ bail-application.ts # ★
■   ■       ■■■■ complaint.ts       # ★
■   ■       ■■■■ research.ts        # ★
■   ■       ■■■■ summarise.ts       # ★
■
■■■■ scripts/

```

---

|                   |                                                         |
|-------------------|---------------------------------------------------------|
| ■ ■■■ migrate.sql | # Schema DDL – paste in Supabase SQL Editor             |
| ■ ■■■ seed.sql    | # 1 lawyer + 2 clients + 3 cases + deadlines + hearings |
| ■                 |                                                         |
| ■■■ presentation/ |                                                         |
| ■■■ Lawris.pptx   | # 6-slide pitch deck (16:9)                             |
| ■■■ CUE_CARD.md   | # Printable cheat sheet                                 |
| ■■■ build_pptx.py | # python-pptx generator (re-runnable)                   |

★ = files where most of the demo quality lives.

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## 10. End-to-End Demo Path (call traces)

### 10.1 "Open the POCSO case → see the critical deadline"

```
Browser GET /cases/aaaaaaaa-...
→ src/app/cases/[id]/page.tsx (server component)
→ loadCase() (parallel Supabase queries: case, deadlines, documents, hearings, research)
→ Supabase REST (service-role, filtered by lawyer_id)
→ renders <CaseDetail>
→ src/components/case-detail.tsx (client)
→ tab "Deadlines" → <DeadlineList>
→ src/components/deadline-list.tsx
→ classifyUrgency(due_date) from src/lib/deadlines.ts
→ "in 2d" pill rendered red via urgencyClass()
```

### 10.2 "Click 'Draft Bail Application' → watch it stream"

```
User clicks button in <AiDraftPanel>
→ fetch POST /api/ai/draft { case_id, doc_type: "bail_app" }
→ src/app/api/ai/draft/route.ts
→ zod parse body
→ Supabase: load case + client + lawyer rows
→ bailApplicationUserPrompt(...) – JSON-stringify case data
→ streamDraft() → gemini.models.generateContentStream({...})
→ for-await chunks → encode → ReadableStream controller.enqueue
→ on stream end: insert into documents (source: ai_drafted, content: collected)
→ Browser reads stream chunk-by-chunk
→ setDraft(d => d + decodedText) – re-renders editor pane each tick
→ mdToHtml() converts incremental Markdown to HTML
```

### 10.3 "Add a hearing → AI summary updates"

```
User submits hearing form in <HearingPanel>
→ fetch POST /api/cases/:id/hearings
→ zod parse
→ Supabase insert hearing_logs row
→ if next_date: Supabase insert deadlines row (auto)
→ Supabase fetch case + all hearings
→ generateText() with summarise prompt → 5-sentence brief
→ Supabase update cases.ai_summary
→ router.refresh() in client → server component re-fetches → header re-renders with new summary
```

### 10.4 "Ask research → get cited answer"

```
User types question in <ResearchPanel>
→ fetch POST /api/ai/research { case_id, query }
→ Supabase load case (for ai_summary + case_type context)
→ generateText() with research prompt + responseSchema (JSON forced)
→ JSON.parse the response → ResearchResponse
→ Supabase insert research_notes (Q + A + citations stored)
→ Browser renders structured turn (answer + cited cases + statutes + follow-ups)
```

# 11. Environment Setup

## 11.1 .env.local (already configured)

```
GEMINI_API_KEY=AIZA... # AI Studio key (rotate post-hackathon)
GEMINI_MODEL_DRAFT=gemini-2.5-flash # bail/plaint
GEMINI_MODEL_RESEARCH=gemini-2.5-flash-lite # JSON Q&A
GEMINI_MODEL_SUMMARISE=gemini-2.5-flash-lite # case brief
NEXT_PUBLIC_SUPABASE_URL=https://rxqhbmtvgdgkubnetpsy.supabase.co/
SUPABASE_SERVICE_ROLE_KEY=eyJ... # NEVER expose to client
NEXT_PUBLIC_SUPABASE_ANON_KEY=eyJ...
LAWYER_ID=11111111-1111-1111-1111-111111111111 # the seeded Priya Mehta UUID
```

## 11.2 First-time setup commands

```
# Node 20 via nvm (already installed)
export PATH="$HOME/.nvm/versions/node/v20.20.2/bin:$PATH"

# Install
cd ~/vit_hackathon
npm install

# Apply schema (one time)
# 1. Open Supabase SQL Editor
# 2. Paste scripts/migrate.sql → Run
# 3. New query → paste scripts/seed.sql → Run

# Run dev server (port 4000 to avoid conflicts)
nohup env PORT=4000 npm run dev > /tmp/lawris-dev.log 2>&1 &
disown

# Verify
curl http://localhost:4000/api/health
```

## 11.3 Demo URLs

| URL                                                                   | What                           |
|-----------------------------------------------------------------------|--------------------------------|
| http://10.11.11.182:4000                                              | Dashboard (works on same LAN)  |
| http://10.11.11.182:4000/cases/aaaaaaaa-aaa-a-aaaa-aaaa-aaaaaaaaaaaaa | POCSO case — the demo headline |
| http://10.11.11.182:4000/calendar                                     | Global deadline view           |

## 12. Production-Build Sanity

next build is green — verified output:

| Route (app)                      | Size    | First Load JS |
|----------------------------------|---------|---------------|
| ■ f /                            | 860 B   | 96.8 kB       |
| ■ f /api/ai/draft                | 0 B     | 0 B           |
| ■ f /api/ai/research             | 0 B     | 0 B           |
| ■ f /api/cases                   | 0 B     | 0 B           |
| ■ f /api/cases/[id]              | 0 B     | 0 B           |
| ■ f /api/cases/[id]/hearings     | 0 B     | 0 B           |
| ■ f /api/deadlines/[id]/complete | 0 B     | 0 B           |
| ■ f /api/deadlines/upcoming      | 0 B     | 0 B           |
| ■ f /api/health                  | 0 B     | 0 B           |
| ■ f /calendar                    | 175 B   | 96.1 kB       |
| ■ f /cases                       | 860 B   | 96.8 kB       |
| ■ f /cases/[id]                  | 9.11 kB | 96.4 kB       |
| ■ ■ /cases/new                   | 2.02 kB | 89.3 kB       |

Total client bundle ~96 KB shared. No type errors, no lint errors.

## 13. What's Built vs Future Scope

### ■ Built (in repo, working live)

- 7-table relational schema with FKs and enums
- Single-lawyer auth scaffold (env-pinned)
- Case CRUD: list, detail, create form
- Auto-computed chargesheet deadlines (60/90-day rule from FIR data)
- Color-coded urgency on dashboard, case detail, calendar
- Streaming AI bail-application drafter
- Streaming AI plaint drafter
- Hearing logging with auto-generated next-date deadline
- AI summary regeneration after every hearing
- Case-grounded research with structured JSON citations
- Document panel with expandable Markdown viewer
- Markdown-to-HTML renderer for streamed content
- Pitch deck ( [presentation/Lawris.pptx](#) ) and cue card

### ■ Future scope (explicit cuts; documented for Round 2)

- Multi-user auth + Supabase RLS
- Document upload + PDF/image storage to Supabase Storage
- OCR on uploaded scans
- IndianKanoon API integration for verifiable citations
- Phase-aware document suggestions ("you're in pretrial → draft bail app?")
- Vakalatnama, written statement, legal notice templates
- Hindi & Marathi UI
- Mobile (React Native or PWA)
- Email/SMS deadline reminders
- DOCX/PDF export (currently relies on browser Print)
- Document comparison & redline diffing
- Voice-to-hearing-log
- Bench-aware outcome prediction
- Vercel deployment (currently on local VM)

## 14. Known Limitations (acknowledge proactively to judges)

| Limitation                      | Mitigation strategy                                                                                                 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------|
| AI may fabricate citations      | Prompt explicitly says "OMIT if not confident — do not fabricate." Real fix: IndianKanoon API grounding (post-MVP). |
| No RLS — single-user MVP        | Acceptable for hackathon; pinned to seeded lawyer via <code>LAWYER_ID</code> . RLS is a 1-day post-MVP task.        |
| No real authentication          | Same scoping — Supabase Auth + RLS is the production answer.                                                        |
| Streaming only on draft route   | Research and summarise routes are single-shot for JSON parsing reliability.                                         |
| TipTap installed but unused     | Custom Markdown renderer was simpler for streaming compat; TipTap is wired and ready if rich-text editing is added. |
| Aadhar stored plaintext in seed | Seed has placeholder <code>XXXX-XXXX-1234</code> values, not real PII. Production needs KMS-backed encryption.      |
| Disk-tight VM (~200 MB free)    | Caches periodically need clearing ( <code>rm -rf ~/.cache/go-build &amp;&amp; npm cache clean --force</code> ).     |

## 15. Roadmap

### Next 30 days (post-hackathon)

1. **IndianKanoon citation grounding** — every research citation verified against the IndianKanoon API
2. **Document upload + OCR** — Supabase Storage + Tesseract.js or Google Vision
3. **Multi-user auth** — Supabase Auth, full RLS, lawyer firm hierarchy
4. **Vercel production deployment** with custom domain
5. **DOCX/PDF export** of drafted documents

### Next 90 days

1. Mobile (React Native shell sharing the API)
2. Hindi + Marathi UI and prompt translations
3. Voice-to-hearing-log (Whisper API → existing hearing endpoint)
4. Bench-aware outcome prediction (analyze each judge's grant/deny patterns)
5. Team collaboration features (shared cases, comments, approval workflows)

### Market path

- **Beachhead:** Solo and 2–10 partner firms in Pune/Mumbai/Bangalore
- **Pricing:** ₹500/lawyer/month
- **TAM math:** 1.4M registered advocates × 1% adoption × ₹500 × 12 = **₹84 cr ARR**
- **Distribution:** Bar Council partnerships, law-college tie-ins, content marketing on legal blogs

# 16. Operational Runbook (when something breaks)

| Symptom                                                     | Diagnosis                                                            | Fix                                                                                                 |
|-------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Dashboard shows "0 critical · 0 high"                       | Seed dates are stale — <code>current_date - X days</code> in old run | Re-run <code>scripts/seed.sql</code> to re-anchor dates                                             |
| <code>/api/health</code> returns 500                        | Env var missing                                                      | Check <code>.env.local</code> ; restart dev server ( <code>kill -f "next dev"</code> → re-launch)   |
| <code>/api/ai/draft</code> errors instantly                 | Gemini API key invalid or rate-limited                               | Verify <code>GEMINI_API_KEY</code> ; check AI Studio dashboard for quota                            |
| Stream cuts off mid-document                                | Network blip; rare                                                   | Click Draft again — second attempt usually works                                                    |
| <code>next build</code> fails with disk error               | VM disk full                                                         | <code>rm -rf ~/.cache/go-build &amp;&amp; npm cache clean --force</code> (frees ~800 MB)            |
| Browser can't reach <code>localhost:4000</code> from laptop | VS Code tunnel broken                                                | Use <code>http://10.11.11.182:4000</code> directly (same-LAN)                                       |
| Dev server SIGTERMs randomly                                | Background-task lifecycle                                            | Use <code>nohup env PORT=4000 npm run dev &gt; /tmp/lawris-dev.log 2&gt;&amp;1 &amp;; disown</code> |

# 17. Round 1 Pitch Materials (already shipped)

Located in presentation/:

| File                 | Use                                                                                                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Lawris.pptx</u>   | 6 slides, 16:9, indigo brand. Title → Problem → Solution → Demo placeholder → Competitor matrix → Roadmap+Ask. Speaker notes on every slide with the verbatim lines and demo timings. |
| <u>CUE_CARD.md</u>   | Printable A4 cheat sheet — cold open, demo clicks-with-timestamps, Q&A one-liners, pre-flight checklist, forbidden phrases.                                                           |
| <u>build_pptx.py</u> | python-pptx generator. Re-run <code>python3 build_pptx.py</code> after editing copy/colors to regenerate the deck.                                                                    |

Pitch shape: 5 minutes total, slides + live demo, generalist judges. Demo is the centerpiece (135 seconds inside the pitch).



## 18. Verification Checklist (before any demo)

Run this twice; failing any step means demo is not ready:

1. ■ `curl http://localhost:4000/api/health` returns `{"ok":true,"env":{"...all true...}}`
2. ■ Dashboard at `/` shows "1 critical · 1 high · 3 active matters"
3. ■ POCSO case ( `/cases/aaaaaaaa...` ) shows chargesheet deadline in **red** with "in 2d" or sooner
4. ■ Click "Draft Bail Application" — first character appears within **5 seconds**, full document within **35 seconds**
5. ■ Drafted bail app cites *Satender Kumar Antil*, mentions Article 21, includes the FIR number and accused name
6. ■ Adding a hearing log triggers visible AI summary update on case header within ~5 seconds of save
7. ■ Research panel question returns at least one cited Supreme Court case
8. ■ `/calendar` page groups deadlines by urgency, color-coded
9. ■ Backup demo video plays in one click (record before pitching)
10. ■ Print-out of one bail application output in hand for the judges

If all 10 pass, you are demo-ready.